CLASS TEST ASSESSMENT TOOL - 1	NAME : TANUSHRI S.V. REG NO: 192321163 CODE : ITA0502 COURSE: COMPUTER VISION.

1. IMAGE ENHANCEMENT IN SPATIAL DOMAIN.

DEFINITION:

* Image enhancement in the spatial domain improves image quality by directly modifying pixel values.

a) Gray - Level Transformations.

Modify pixel intensity values.

Techniques: Negative, Log, Power-law [Gamma] Contrast Stretching.

Use: Improves brightness and contrast.

b) Histogram Processing.

Improves image contrast by redistributing gray levels.

Techniques: Histogram Equalization, Histogram Specification.

Use: Enhances low - contrast images.

c) Spatial filtering.

Uses a mask [Kernel] to process neighboring pixels.

Smoothing Filters: Mean, Gaussian, Median [noise removal].

Sharpening Filters: Laplacian, Sobel, unsharp Masking [edge enhancement].

Justification:

Gray-level transformation → Brightness adjustment.

Histogram processing → Better Contrast.

Spatial filtering → Noise removal and Sharpen edges.

Application:

* Medical imaging, Satellite images, CCTV Surveillance, Document enhancement.

Conclusion:

* Spatial domain enhancement improves image quality by enhancing brightness, Contrast, reducing noise, and highlighting important details.

2. IMAGE SMOOTHING IMAGE VS IMAGE SHARPENING.

* Image Smoothing Image Sharpening removes noise enhances edges Blurs image makes clearer uses Low-pass filters uses High-Pass filters.

* Removes high-frequency components enhances high-frequency components.

Image Smoothing.

* Smoothing reduces noise and unwanted details. It uses low-pass filters.

Working:

* A filter mask is moved over the image.

* Pixel values are calculated using neighbouring pixels.

* This reduces sudden intensity changes.

Eg: Mean filter, Median, filter.

Application: Noise removal, medical images preprocessing.

IMAGE SHARPENING:

* Sharpening enhances edges and fine details. It uses high-pass filters.

Working:

* Detects sudden changes in intensity.
* enhances edges and boundaries.
* Makes the image appear clearer.

Eg: Laplacian, Sobel, Prewitt.

CONCLUSION:

* Smoothing is used when noise needs to be removed, while sharpening is used when edges and details need to be enhanced.

3. IMAGE ENHANCEMENT IN FREQUENCY DOMAIN

INTRODUCTION:

* Frequency-domain enhancement processes an image according to its frequency components.

PROCESS:

Image
↓
Fourier Transform
↓
Frequency Domain
↓
Apply Filter.
↓
Inverse Fourier Transform.
↓
Enhanced Image.

LOW - PASS FILTER:

* A low - pass filter allows low frequencies and reduces high frequencies.

EFFECTS:

* Removes noise.
* Reduces fine details.
* Smooths the image.
* Can cause blurring.

APPLICATIONS:

* Noise removal, image smoothing. Preprocessing.

HIGH - PASS FILTER:

* A high - pass filter allows high frequencies and reduces low frequencies.

EFFECTS:

* Enhances edges.
* Highlights fine details.
* Sharpens the image.

APPLICATIONS:

* Edge enhancement, sharpening defect detection.

CONCLUSION:

* Low - pass filters are suitable for smoothing, while high - pass filters are suitable for sharpening & edge enhancement.

4. IMAGE SEGMENTATION:

DEFINITION:

* Image Segmentation is the process of dividing an image into meaningful regions or Objects.

TYPES:

1. Point Detection: Detects isolated points
2. Line Detection: Detects lines.
3. Edge detection: Detects Object boundaries.
4. Thresholding: Separates Objects based on intensity.
5. Region-based: Groups similar neighboring pixels.

SUITABLE TECHNIQUE:

* For object and defect detection, edge detection with thresholding is Suitable.

Reason: Edge detection identifies the Object's boundary, while thresholding Separates the Object from the background.

CONCLUSION:

* Segmentation helps in Object recognition, medical imaging, defect detection, and image analysis.

5. EDGE DETECTION, EDGE LINKING AND BOUNDARY DETECTION

INTRODUCTION:

* Edges are important features that represent sudden changes in intensity. Edge detection, linking, and boundary detection are important for recognizing objects and defects.

1. EDGE DETECTION:

* Edge Detection identifies sudden intensity changes in an image.
Eg: Sobel, Prewitt, Roberts, Canny.

Dark Region | Bright Region.
↑
Edge.

USES: Object recognition, Segmentation, defect detection.

2. EDGE LINKING:

* Edge linking connects broken or disconnected edge pixels.

Broken edges → Edge Linking → Continuous edge.

* It uses neighbouring pixels, direction and intensity to connect edges.

3. BOUNDARY DETECTION:

* Boundary detection identifies the complete outline of an object.

Image

↓

Edge Detection

↓

Edge Linking

↓

Boundary Detection

↓

Object Shape.

CONCLUSION:

* Edge detection finds edges, edge linking connects them, and boundary detection forms the complete object boundary. Together, they improve object recognition and defect detection.